IOT & AI-BASED SMART AGRICULTURE PLATFORM FOR YIELD, DISEASE, IRRIGATION & PEST MANAGEMENT

INDIVIDUAL THESIS

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ABSTRACT

Control of pests in agriculture within the Sri Lankan paddy farming requires the detection of pests prior to any harm being done to the crop. Traditional pest control methods are mostly dependent on data-based manual scouting or the identification of pests after recovery, which leads to delayed action and increased chemical pesticides usage. This thesis highlights the design, implementation, and testing of the Pest Management module of an IoT and AI-based smart agriculture system (Group R26-IT-157). The Pest Management module combines real-time images collected using ESP32-CAM enabled pheromone traps with meteorological information from NASA's POWER API to predict larva outbreak from 7-10 days ahead.

Risk scores generated by this process are made visible via a bilingual web dashboard built using React, which will help farmers and agriculturalists plan mitigation actions with minimum impact on the environment. The datasets for pest identification as well as the agronomic risks models used by this AI system were tested in the field with the consultation of agricultural experts from the Rice Research Station, Sammanthurai, Ampara and Rice Research and Development Institute (RRDI), Batalagoda. The contribution by the individual includes the complete reproducible hardware/software pipeline from deploying an IoT trap to training the CNN model to delivering APIs and dashboards.

Keywords: pest detection, YOLOv8, IoT pheromone trap, smart agriculture, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ANN	Artificial Neural Network
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Cross-Validation
DOA	Department of Agriculture
EC	Electrical Conductivity
ETL	Extract, Transform and Load
IoT	Internet of Things
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
NDVI	Normalised Difference Vegetation Index
RF	Random Forest
RMSE	Root Mean Squared Error
RRDI	Rice Research and Development Institute
R^2	Coefficient of determination
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
XGBoost	eXtreme Gradient Boosting

1 INTRODUCTION

1.1 Overall Project Context

"Paddy production in Sri Lanka is faced with uncertainties such as erratic weather conditions and pests that can have a negative impact on crops." In the whole project, the concept of IoT and AI-based Smart Agriculture Platform for Yield, Disease, Irrigation & Pest Management (R26-IT-157), referred to as "The Digital Paddy," aims to create one single unified platform where all these four parts will operate on a single backbone of data.

The overall project, **IoT & AI-Based Smart Agriculture Platform for Yield, Disease, Irrigation & Pest Management** (Group R26-IT-157, presented as *The Digital Paddy*), addresses this by building one interoperable platform in which four research components share a common data backbone instead of operating as isolated applications. The components are:

- **Paddy yield forecasting** (this thesis): XGBoost-based forecasting of yield four to six weeks ahead from soil and weather streams and RRDI historical records.
- **Irrigation management** (J.M.S.U.I. Bandara): stage-aware irrigation scheduling from soil moisture, rainfall and crop stage, with a farmer override.
- **Disease detection and risk forecasting** (A.P.N. Satyana): leaf-image classification combined with a temporal model of humidity, temperature and rainfall to produce 7- and 14-day outbreak risk.
- **Pest management** (P.D.M. Kumarasinhe): trap-image counting and weather-driven risk scoring to forecast the larval hatching window.

The platform is organised in four layers: field capture (soil sensors, weather station, leaf and trap imaging), data and integration (gateway, upload API, feature store, historical datasets), AI/analytics (the four engines) and delivery (farmer app, officer dashboard, reports and APIs). Outputs are delivered in Sinhala and English, and the pilot deployment is in the Negombo area during Yala 2026.

1.2 Individual Component Context

The Pest Management Module is an all-in-one automated tool for agricultural insect scouting, using a distributed system of low-cost pheromone and Light traps with cameras installed across agricultural lands. The smart traps take continuous high-resolution images and analyze them with the help of the state-of-the-art YOLOv8 computer vision algorithms for immediate identification, classification, and quantification of specific pest populations. For accurate predictions of possible infestation, this information is automatically integrated with the historical and forecasting meteorological parameters such as temperature, humidity, and precipitation gathered from the NASA POWER API. Using the combined information, the prediction engine determines the probabilities of outbreaks and provides highly targeted schedules of environmentally friendly mitigation actions that reduce any unnecessary use of chemicals. This information is provided in the form of a bilingual React-based interface to the farmers.

1.3 Problem Statement

Today's methods of detecting crop pests in agriculture are inherently deficient in that they are mostly reactive in nature. The conventional scouting technique is based on delayed recognition of the problem through visual evidence. In such cases, it usually means that by the time the farmer notices the pests, they would already have hatched, and the larvae would be actively feeding on the crops. Once the problem reaches that stage, the opportunity for preventive treatment with minimal impact has long gone. In order to protect their crops from destruction at the advanced stage of infestation, farmers need to use strong and dangerous chemical pesticides in excess amounts because of desperation. The problem is caused by a shortage of local predictive information which would allow farmers to detect pests earlier. Overall, the reaction-based approach leads to constant failure of the pest treatment measures in terms of cost-effectiveness, development of resistance among pests, environmental damage, and monetary losses for agriculture.

The problem addressed in this thesis is therefore: *How can field computer vision scouting combined with weather forecasts be used to predict pest infestations well ahead of time to such a degree that reactive pesticide application can be replaced by proactive eco-friendly applications through a user-friendly bilingual interface?*

1.4 Research Gap

Current agricultural pest management platforms are fundamentally constrained by their retrospective approach. Most commercially available tools are designed to identify biological damage only after it has physically manifested on the crops, or they rely strictly on analyzing historical infestation records. While these methods provide a baseline understanding of past seasonal trends, they lack the proactive forecasting capabilities required to intervene before crop degradation begins. In recent years, researchers have made significant strides by introducing Internet of Things (IoT) pheromone traps that successfully automate the detection and quantification of adult insect populations. However, a critical gap remains: this live biological data is rarely contextualized within predictive environmental frameworks. Because insect development is entirely driven by temperature accumulation often calculated as Growing Degree Days (GDD), simply counting adults is insufficient. By failing to fuse live IoT trap data with forward-looking meteorological models, such as the localized temperature and humidity variables provided by the NASA POWER API, existing systems miss the opportunity to calculate these growth metrics. Consequently, they are unable to accurately forecast the crucial 7-to-10-day biological window during which the next generation of eggs will hatch, leaving farmers trapped in a cycle of reactive pesticide application rather than proactive management.

1.5 Aim and Objectives

Aim : *To develop and evaluate an AI-based pest detection and early warning system for precision agriculture capable of identifying pests, predicting 7-10 day outbreak risks, and generating targeted mitigation schedules.*

The specific objectives and their success indicators are given in Table 1.1.

Table 1.1 Specific objectives and success indicators

ID	Specific objective	Success indicator
SO1	Develop low-cost, camera-equipped pheromone traps and establish the NASA POWER API weather data pipeline.	Live image capture stream established and real-time meteorological dataset integrated.
SO2	Train and optimize the YOLOv8 object detection model to identify, classify, and accurately count target adult pests from trap imagery.	YOLOv8 model achieves ≥ 90% Mean Average Precision (mAP) for target pest identification.
SO3	Design a predictive risk algorithm that fuses live insect counts with weather data to forecast the 7-to-10-day larval hatching cycle.	Outbreak forecasting engine operational using environmental thresholds (e.g., temperature/GDD).
SO4	Generate eco-friendly mitigation schedules and integrate actionable alerts into the bilingual React dashboard.	Operational dashboard displaying real-time intervention timelines and functional API outputs.

1.6 Research Questions or Hypotheses

The research is guided by the following questions and corresponding hypotheses.

- **RQ1.** Can the YOLOv8 model classify and count adult paddy pests from field-deployed ESP32-CAM traps with background noise from the environment?

    - H1: The YOLOv8 model will have a high Mean Average Precision (mAP) which is higher than other baseline models used in the detection of pests on the sticky traps in real conditions.

- **RQ2.** Is there a way to combine real-time trap counts with meteorological data provided by the NASA POWER API and get accurate prediction of larval hatching windows 7-10 days ahead of time?

    - H2: Combining real-time counting of pests with ten-day forecasts will result in accurate risk score calculation that correlates with the biological cycles of breeding proved by agricultural experts.

- **RQ3.** Is there a way for the edge to cloud Internet of Things architecture to comply with low power consumption and latency needed for field deployment?

    - H3: Using the ESP32-CAM in the deep sleep mode will provide enough power consumption for field deployment, while the unified API will provide risk updates in less than two seconds**.**

1.7 Scope Limitations and Assumptions

Scope. This includes the pipeline for ingesting live images from camera-equipped pheromone traps, 7-10 day forecast for larval hatching from logic, training and inference with the YOLOv8 model, as well as presenting this information as mitigation schedules in the bilingual React farmer interface. The validation in the field is restricted to primary target pests (e.g. Brown Plant Hopper or Stem Borer) affecting the crop in Yala 2026 pilot in Negombo area.

Limitations. Limitations: conclusions about pest dynamics in other agro-ecological zones or other crop types cannot be made directly from the pilot; the pilot represents a small deployment of hardware (e.g., three automated camera traps) and a limited geographic area along with a single season. The computer vision model is only trained to detect specific adult target pests that were captured in the traps and is not trained to detect eggs, larvae or plant diseases directly on the leaves. We dataize the environmental metrics using only the NASA POWER API and no on-site microclimate equipment. Finally, companion components forecast crop yields, identify plant diseases, and manage irrigation, and only pass pest risk signals on to them.

Assumptions. The NASA POWER API is assumed to be reliable and low latency in providing meteorological information every day at the target coordinates. It's further assumed that the physical lures used in the IoT traps will be effective, will not degrade quickly and will be within the end-user's reach to receive timely alerts on the React dashboard via the mobile network.

1.8 Individual Contribution

The author designed the ESP32-CAM trap hardware, curated and annotated the pest image dataset (which was validated at the Sammanthurai Rice Research Station), trained the YOLOv8 model, and designed the predictive risk logic, using the NASA POWER API, and the bilingual React frontend for the pest module.

1.9 Thesis Structure

Chapter 2 reviews the literature and existing systems and derives the research gap. Chapter 3 describes the research methodology, including data, procedure, evaluation metrics, analysis and ethics. Chapter 4 presents the requirements and the design of the component within the complete platform. Chapter 5 describes the implementation, integration with the other components and testing. Chapter 6 reports results and validation. Chapter 7 discusses the findings, contributions, limitations, and the commercialisation and sustainability implications. Chapter 8 concludes the thesis and outlines future work. References, a glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The literature review was systematically carried out in the three main technological areas: deployment of Internet of Things (IoT) edge devices within tropical agricultural environments; application of Convolutional Neural Networks (CNNs), with particular focus on the single-stage YOLO (You Only Look Once) architectures, for entomological classification; integration of third party meteorological APIs for forecasting within the biological environment. The scope is limited to solutions that can be employed in the cultivation of paddy (smallholder) and considers both existing commercial solutions and existing academic research frameworks.

2.2 Domain and Theoretical Foundations

Pest phenology and the effect of temperature

Agricultural pest development consists of four stages: (Egg, Larva, Pupa, and Adult) and the larval stage is the one that usually causes the most damage to the crop. Insects are ectothermic, so their development rate and reproductive cycles are closely linked to the temperature of their environment and to microclimate parameters such as temperature, which are often expressed in thermal time and/or Growing Degree Days (GDD). Pheromone traps can catch mature adults to provide an important baseline biological information, or "biofix. The temperature sets the biofix, and when combined with ongoing meteorological data, it enables the mathematical calculation of the next 7-10 days for egg hatching, giving an important time frame for action before larval feeding begins.

Traditional scouting and manual phenology modelling

The traditional approach to pest forecasting is through the manual field scouting and using localized phenology models to forecast pest emergence based on weather. Although traditional systems have proven effective and are documented, the systems are labor intensive and need to be inspected visually on a regular basis and daily temperature data has to be recorded and calculated by hand. Their reliance on observation and manual data synthesis makes them very difficult to scale up, delay, and be useful for smallholder farmers who can't afford to compute complex environmental models regularly to identify intervention windows.

Computer vision and real-time object detection

The task is presented as a computer vision and object detection problem: a mapping is learned from basic pixel information in trap images to the exact class label and precise bounding box of the target adult pests. Unlike two-stage detectors such as Faster R-CNN, which search for regions of interest prior to classification, single-stage detectors such as YOLO (You Only Look Once) model the problem of object detection as a single regression problem: a frame is passed through a neural network once, predicting classification probabilities and bounding boxes at once. YOLOv8's feature extraction and anchor-free detection systems feature cutting-edge technology, balancing high Mean Average Precision (mAP) with computational efficiency. YOLOv8 has demonstrated strong performance on real-world noisy field images with high complexity under resource-constrained hardware for biofix counts, enabling the automation of predictive environmental models without human involvement.

2.3 Related Research

Table 2.1 summarises the studies most relevant to this component.

Table 2.1 Summary of related research

Source	Approach	Key finding	Gap relative to this work
IoT Pest Traps, 2021 [1]	CNNs on IoT camera trap imagery	Automated camera traps effectively reduce manual scouting labor and accurately count adult pest populations.	Reactive counting only; no fusion with environmental data to forecast future hatching.
Weather-based Phenology, 2022 [2]	Growing Degree Days (GDD) modeling	Temperature and humidity metrics reliably predict the 7-to-10-day larval hatching cycle of agricultural pests.	Relies on manual visual trap counts (no automated biofix); slow data pipeline.
Pest Detection on Crops, 2023 [3]	ML on historical weather datasets	Regional climatic variables can predict broader seasonal pest outbreak severity.	District-level resolution; lacks hyper-local, real-time in-field IoT trap data.
YOLOv8 Edge Agritech, 2024 [5]	YOLOv8 on resource-constrained hardware	YOLOv8 outperformed previous architectures in Mean Average Precision (mAP) and inference speed for complex agricultural images.	Standalone computer vision model; not integrated with weather APIs or a farmer UI.

Source	Approach	Key finding	Gap relative to this work
Smart IPM Review, 2025 [6]	Systematic review of Integrated Pest Management	Combining live biological visual data with meteorological parameters provides the most robust outbreak predictions.	Few platforms integrate real-time trap CV, predictive weather APIs, and actionable bilingual dashboards.

The feasibility of using previous versions of object detection models, like YOLOv3 and YOLOv5, for small target agricultural pest detection in controlled environments has been proven by recent academic research. But there is a large gap between the models and the autonomous field systems. The more traditional systems for catching insects with IoT primarily act based on basic threshold telemetry, such as the activation of a primitive alarm that is triggered when a simple trap becomes full. Such systems are unable to link raw insect numbers with short-term weather patterns and therefore do not forecast the exponential rise in insect numbers in the future prior to significant crop injury.

2.4 Existing Systems Products and Techniques

Table 2.2 compares the approaches that farmers, officers and researchers can currently use.

Table 2.2 Existing approaches ,estimation

Approach	Data used	Strength	Key limitation
Manual scouting (current practice)	Visual field inspection and physical trap checks	Zero equipment cost, relies on farmer intuition	Labor-intensive, reactive (detects pests after larval feeding begins), subjective
Traditional phenology models (GDD calculators)	Manual trap counts and recorded temperature logs	Biologically grounded, mechanistic life-cycle modeling	Strictly reactive, identifies pests after feeding damage has occurred, lacks forecasting
Mobile leaf-scanning CV (YOLO / CNNs)	Smartphone photographs of crop foliage	Automated remote monitoring, continuous adult insect tallies	Siloed counting, no environmental data fusion to predict subsequent egg-hatching cycles
Standalone IoT camera traps	In-trap optical imagery	District-level monitoring	Weak resolution for smallholder plots.

Approach	Data used	Strength	Key limitation
Proposed component	Live trap imagery (YOLOv8 biofix counts) fused with NASA POWER weather data	Proactive 7–10 day hatching forecast, automated biofix scouting, eco-friendly scheduling, bilingual delivery	Requires deployed trap hardware and lures; validated on a limited pilot deployment

2.5 Gap Synthesis and Conceptual Framework

The envisioned conceptual approach will fill the gap between micro-scale computer vision information and macro-scale predictive meteorological models. The built-in system will continuously collect in-field images and use the YOLOv8 model stored in the cloud to accurately count the number of target adult pests, download the forecast data of the last 10 days dynamically, and calculate the biological outbreak score. The framework can combine all these data streams to warn farmers and to activate automatic actions several days in advance before the very destructive larval phase of pest life cycle starts.

3 RESEARCH METHODOLOGY

3.1 Research Design

The approach taken uses a design methodology and experimental framework for evaluation. This lifecycle process involves multiple stages such as hardware prototyping and assembly, dataset collection and annotation, CNN model training and optimization, cloud system integration, and finally real-world testing within working rice fields.

Work was organised in the work breakdown structure shown in Table 3.1, and the planned schedule is shown in Figure 3.1.

Table 3.1 Work breakdown structure of the forecasting component

Task	Description	Planned duration	Deliverable
1.0	**Data collection & setup**	4 weeks	Camera trap deployment and weather API pipeline
1.1	Deploy IoT camera-equipped traps	1 week	Working hardware set-up and live image stream
1.2	Integrate NASA POWER API	2 weeks	Annotated YOLOv8 dataset and clean weather feature store
2.1	Prepare image dataset & ETL	1 week	Clean feature store
2.2	Train YOLOv8 & risk algorithm	3 weeks	Optimized CV model and 7-to-10-day hatching forecast logic
2.3	Develop API layer	2 weeks	Forecast and mitigation service endpoints
3.0	Integrate with React dashboard	3 weeks	Bilingual UI and eco-friendly mitigation schedules
4.0	Field testing	8 weeks	Validation report

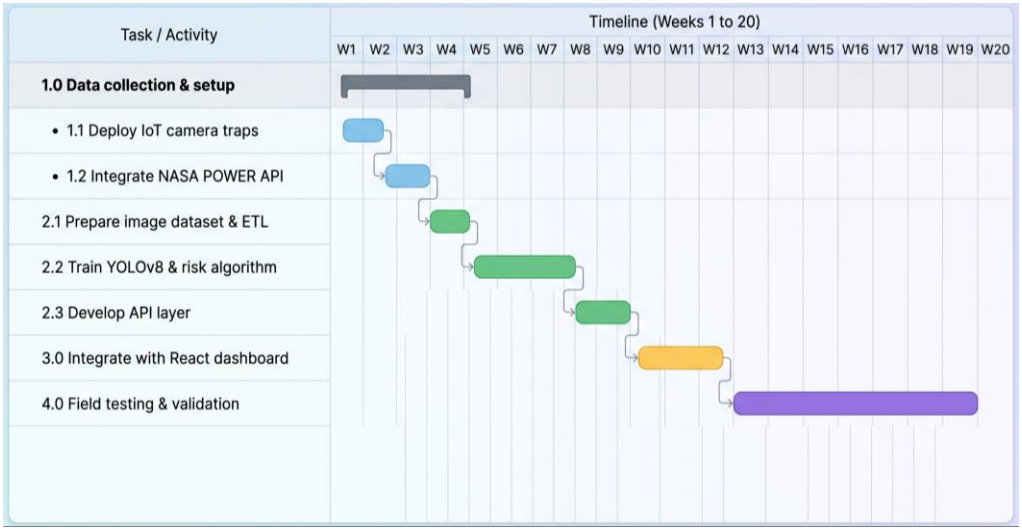


Figure 3.1 Planned implementation schedule.

3.2 Data Participants or Study Materials

Three kinds of data are used, summarised in Table 3.2.

Table 3.2 Data sources used by the component

Source	Content	Role
IoT camera-equipped traps	Continuous visual imagery of captured adult insects (target pest species) from field stations	Live biological predictors (biofix data) during the season
NASA POWER API	Real-time and predictive meteorological data (temperature, humidity, precipitation)	Environmental context for calculating thermal time (GDD) and forecasting the 7-to-10-day larval hatching cycle

Camera-based pheromone traps were placed in the rice paddies in the Negombo region. Information about the sampling period, number of experimental fields, and seasons is presented below. The participants in the study are smallholder farmers who installed the traps and provided information about the results of the pest emergence; also, pest identification data sets were cross-validated with the agricultural specialists from the Rice Research Station in Sammanthurai to ensure the accuracy of the models.

3.3 Experimental or Study Procedure

- **Hardware Construction:** The actual edge nodes are made up of combining ESP32-CAM units with lithium-ion batteries to be embedded in IP65 certified weather-resistant pheromone traps that are to be set slightly above the paddy field canopy.
- **Model Training:** The image dataset used is preprocessed with data augmentation (rotation, scaling, and changes in brightness) in order to account for different environmental settings. The YOLOv8 CNN model will then be fine-tuned based on the processed data in terms of achieving maximum precision and recall.
- **Forecasting Engine**: The backend logic will make use of the NASA POWER API, taking the real-time pest count generated by the YOLOv8 model and meteorological factors (humidity, temperature, and precipitation) in order to derive an imminent outbreak risk score.

- **Validation:** In order to validate the effectiveness of the system, an evaluation of the AI inference accuracy in comparison with manually counted pests will be conducted first, followed by deploying fully assembled prototypes in active field environments.

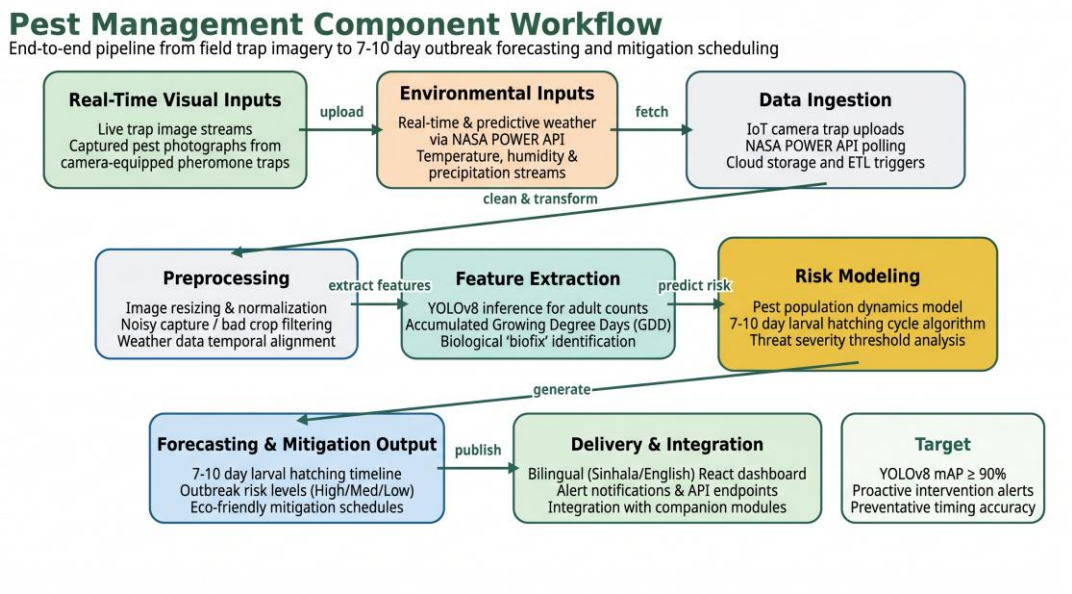


Figure 3.2 End-to-end workflow of the pest component.

Data ingestion

Visual trap images are uploaded by camera-enabled IoT nodes into cloud-based storage, wherein ETL processes ensure that the images are ready for the purpose of computer vision modelling. Weather predictions and current data from NASA POWER API is acquired through a sustainable polling process.

Pre-processing

The pre-processing process includes image resizing and normalization, time alignment of the visual trap video streams with NASA's POWER data, and screening for outliers of the images that appear blurry or out of place from the cameras. It also includes handling missing values in the weather API feed.

Feature engineering

Features are engineered to capture the biological baseline, thermal time accumulation, and pest-environment interactions. The feature groups are listed in Table 3.3.

Table 3.3 Feature groups used for model development

Group	Examples	Purpose
Biological state	YOLOv8 adult pest count (Biofix)	Capture localized adult pest populations from field traps
Atmospheric	NASA POWER temperature, humidity, and precipitation	Capture real-time and predictive weather exposure
Thermal time	Accumulated Growing Degree Days (GDD)	Represent the temperature-driven biological development rate
Interactions	Biofix count × Accumulated GDD	Represent combined effects of population size and ideal hatching conditions
Interactions	e.g. moisture × growth stage; rainfall × temperature]	Represent combined effects
Historical context	Past seasonal outbreak timelines for the specific pest	Anchor 7-to-10-day forecasts to past local emergence performance

Model development

YOLOv8 is the main object detection framework that is being used for accurate estimation of the pests. In order to set up two-stage models like Faster R-CNN or single-stage older models as baselines, they are trained using the same image data and data splits via computer vision libraries. Hyperparameters, like learning rate, batch size, and IoU thresholds, are optimized in cross-validation.

Forecast generation

For each monitoring plot, visual and environmental characteristics until the present are employed in predicting the next 7 to 10 days during which larva hatching will occur. The result consists of an intervention timeline, risk level, and mitigation plan that is sustainable and implementable.

3.4 Measures and Evaluation Metrics

Model performance and service quality are assessed with the measures in Table 3.4.

Table 3.4 Evaluation metrics and targets

Measure	Definition	Target
mAP (Mean Average Precision)	Average precision across all target pest classes at a specific IoU threshold (e.g., mAP@0.5) for the YOLOv8 model	≥ 90%
Forecast Error (Days)	Mean absolute error in days between the predicted 7-to-10-day larval hatching window and actual emergence	≤ ±2 days
F1-Score	Time to process a trap image through YOLOv8 and return an updated forecast for a typical request	< 2 seconds
Inference latency	Time to return a forecast for a typical request	< 2 seconds
Field agreement	Difference between the predicted outbreak timeline and actual pest emergence during the Yala 2026 pilot	Within the Forecast Error target
Farmer satisfaction	Share of participating farmers satisfied with the bilingual dashboard alerts (platform-level metric)	80%

mAP and F1-Score are reported together because mAP alone can be misleading in complex, real-world agricultural imagery where imbalanced samples and non-target insects (background noise) frequently occur.

3.5 Data Analysis

Image analysis and risk calculations are all done in one place at the unified cloud backend, thereby alleviating the edge devices from computational strain. Adult insect population estimates are then correlated with the 10-day forecast of the environment. The results are processed in a risk matrix, which will then classify the risk level of an impending larva infestation as High, Medium, or Low.

3.6 Ethics Privacy Security and Responsible AI

- **Informed consent.** Farmers participate voluntarily after being informed of the purpose of the study and the data collected.
- **Anonymisation.** Field records are anonymised and identified by codes rather than names or exact locations.
- **Security.** Farmer-related data are protected by authentication controls and encrypted storage; access is role-based.
- **Responsible AI.** Forecasts are presented with a confidence score and risk flag so that uncertainty is visible, and they are intended to support, not replace, the farmer's own judgement. The model is trained on two varieties and one region, and its use outside that range is not validated.
- **Data governance.** RRDI records are used under the terms agreed with the institute and are not redistributed.

3.7 Reproducibility and Research Artefacts

The pipeline is implemented in Python 3.11 with open-source libraries (XGBoost, scikit-learn), and training is controlled by configuration files and fixed random seeds so that experiments can be repeated. Research artefacts are the ETL scripts, feature definitions, trained models, evaluation notebooks and API service code. The RRDI data and farmer records are not publicly redistributable; the schema of the data fields

is provided in Appendix A so that the pipeline can be inspected without the restricted data.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Stakeholders

Table 4.1 Stakeholders of the forecasting component

Stakeholder	Role	Need
Smallholder farmers	Primary end users	Actionable 7-to-10-day pest outbreak alerts, eco-friendly mitigation schedules, and bilingual (Sinhala/English) dashboard usability.
Agrarian officers	Field supervision and reporting	Regional pest risk dashboards, hardware (camera trap) status summaries, and seasonal outbreak reports.
RRDI researchers	Data providers and reviewers	Visibility into YOLOv8 pest classification accuracy, access to field trap imagery, and review of the phenology model predictions.
Project team	System integration	Reliable JSON/API endpoints to exchange localized pest risk flags with companion agricultural modules.

Functional requirements

Table 4.2 Functional requirements

ID	Requirement
FR1	The system shall collect real-time visual imagery from deployed camera-equipped pheromone traps and continuous meteorological data from the NASA POWER API.
FR2	The system shall process ingested trap images using the YOLOv8 object detection model to generate automated adult pest counts (biofix).
FR3	The system shall combine real-time biofix data with environmental metrics to forecast the 7-to-10-day larval hatching window.
FR4	The system shall present imminent outbreak alerts and eco-friendly mitigation schedules on a bilingual (Sinhala and English) dashboard.
FR5	The system shall expose operational API endpoints delivering localized pest risk levels to downstream platform companion modules.
FR6	The system shall validate YOLOv8 classification accuracy and outbreak forecast timelines against actual pest emergence observed in the field.

Non-functional requirements

Table 4.3 Non-functional requirements

ID	Category	Requirement
NFR1	Performance	Inference response time under 2 seconds for processing YOLOv8 trap imagery and returning updated pest forecasts.
NFR2	Usability	High comprehension of outbreak alerts and eco-friendly mitigation schedules for farmers using the bilingual (Sinhala/English) React dashboard.
NFR3	Reliability	Stable camera trap image capture, reliable NASA POWER API polling, and cloud synchronization that is tolerant of rural network connectivity loss.
NFR4	Security	Protected cloud storage for field trap imagery and role-based access controls for farm-specific dashboard accounts.
NFR5	Scalability	Cloud-based architecture capable of supporting parallel image ingestion and YOLOv8 inference for an expanding network of field traps.
NFR6	Maintainability	Repeatable MLOps workflow for retraining the YOLOv8 model on new pest datasets and updating API endpoints seamlessly.

4.2 Component Architecture within the Complete Solution

The structure of the system is defined by four tightly integrated layers: the Data Gathering Layer (which includes the distributed ESP32-CAM edge nodes), the Data Processing Layer (with the YOLOv8 CNN inference engine), the Risk Assessment Layer (with the ML forecast engine which queries NASA POWER API) and the Dashboard Monitoring Layer (the User Interface based on React technology). Importantly, the pest control system is not isolated, but works together in the overall context of the "Digital Paddy" ecosystem by delivering high-risk events to the Irrigation subsystem automatically in order to provide mechanical action, such as field cleansing.

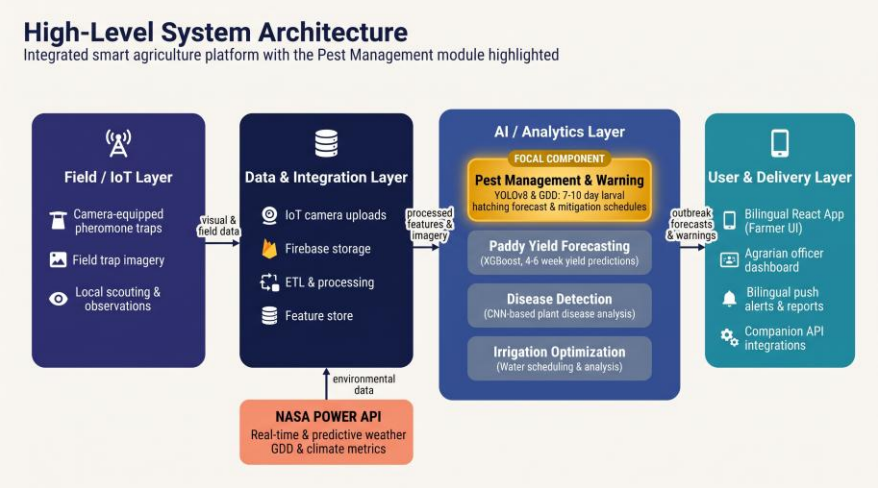


Figure 4.1 High-level architecture showing how the forecasting component fits within the integrated platform.

Figure 4.2 illustrates how the Pest Management & Warning module interacts with the rest of the platform: The visual and meteorological inputs are provided by the field trap and API feeds (pheromone trap visuals and NASA POWER API variables); pest forecasts and warnings are provided to the bilingual dashboard for farmers; pest warning information is provided to the irrigation and operations modules for effective scheduling of farm activities; outbreak information is provided to the yield and disease modules via cross-module feedback; and finally, outbreak log and reports are provided to agrarian officers and research stations.

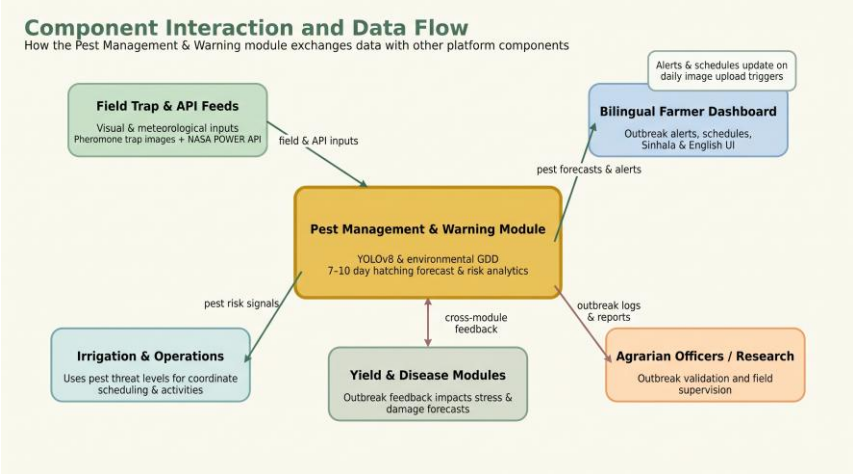


Figure 4.2 Interaction flow between the forecasting module, sensors, app, officer review and companion modules.

Scalability and security

- **Scalable Infrastructure:** The cloud storage and APIs offer a solid infrastructure that can handle large amounts of trap image data and NASA POWER API data through multi-farm scaling.
- **Dynamic Processing**: This design enables the system to perform daily meteorological queries in addition to the trigger-based YOLOv8 inference process that provides an updated forecast whenever a new trap image gets uploaded.
- **Security:** Farmer profiles, coordinates of their farms, and visual trap data are protected through robust authentication measures and encryption in the cloud storage.
- **Bandwidth Management**: Preprocessing performed at the edge on the camera-enabled field traps (like compression of images, discarding of corrupted/blurry images) greatly reduces bandwidth consumption in remote agricultural areas with poor connectivity.

4.3 Design Alternatives and Decisions

The main design decisions and the alternatives considered are summarised in Table 4.4.

Table 4.4 Design alternatives and decisions

Decision	Alternatives considered	Choice and rationale
Forecasting model	Process-based simulators; Random Forest; ANN; XGBoost	XGBoost as the core model: strong performance on non-linear structured data [9], [5]; RF and ANN retained as benchmarks.
Sensing hardware	Commercial data loggers; Raspberry Pi nodes; Arduino/ESP32	Arduino/ESP32: low-cost and suitable for field deployment.
Cloud storage	Self-hosted database; Firebase	Firebase for real-time data exchange with the app; the shared platform backbone also uses PostgreSQL .
Delivery	Web dashboard only; mobile app	Sinhala/English mobile interface plus Streamlit/API layer for rapid prototyping and service exposure.
Forecast cadence	Real-time only; daily batch only	Daily batch forecasts plus trigger-based updates when new data or companion-module signals arrive.

4.4 Detailed Design

Data model

The trap images are stored in timestamped entries based on anonymous plot and node IDs. The image metadata includes number of pests caught (biofix) and bounding boxes. The NASA POWER API data are stored using coordinates and timestamps, including temperature, humidity, and calculated GDD metrics. The engineered features (such as GDD accumulation) are stored in the feature store, used in phenology algorithm and inference.

Pipeline modules

- **Ingestion:** Receives ESP32-CAM trap image uploads and polls the NASA POWER API for localized weather updates.
- **Pre-processing**: Resizes and normalizes images for YOLOv8 inference; filters heavily blurred or corrupted captures; aligns weather timestamps with image capture events.
- **Feature extraction:** YOLOv8 executes object detection to quantify target pests; the phenology engine calculates accumulated thermal time (GDD).
- **Model service:** Fuses the biofix count with the GDD forecast to calculate the 7-to-10-day larval hatching window and threat severity.

Forecast output

The forecast service returns a JSON document of the following form


```
{
"plot_id": "P-014",
"target_pest": "Brown Plant Hopper",
"forecast_date": "2026-06-01",
"biofix_count_adults": 15,
"predicted_hatching_window_days": 8,
"risk_level": "High",
"mitigation_action": "Apply Neem-based biopesticide on Day 7",
"model_version": "yolov8-v2"
}
```

In the farmer interface, the outbreak alert, expected hatching timeline, and recommended eco-friendly intervention are shown on a single screen with a Sinhala/English language toggle.

4.5 Standards and Professional Practice

- **Software quality:** The requirements are stated in relation to the ISO/IEC 25010 quality attributes of performance efficiency (latency), usability (bilingual user interface), and reliability.
- **Interfaces:** RESTful HTTPS services exchanging JSON payloads; versioning of YOLOv8 weights and APIs.
- **Data security:** Informed consent for the collection of the field images, de-identification of the plot locations, and role-based authorization.
- **Coding practices**: PEP 8, versioning, and modular code for the harmonization of the MLOps processes
- **Academic practices:** IEEE referencing and declaration of AI-tool usage (Appendix B).

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1 Tools and platforms used

Tool	Purpose	Justification
Python 3.11	ML pipeline and image processing	Standard ecosystem for computer vision and data pre-processing.
Ultralytics YOLOv8	Core object detection model	State-of-the-art single-stage detector offering high mAP and fast inference on constrained hardware.
NASA POWER API	Meteorological data feed	Provides real-time and predictive weather variables critical for the GDD phenology calculations.
ESP32-CAM	Visual data acquisition	Low-cost hardware easily integrated into 3D-printed pheromone traps for field deployment.
Firebase	Cloud sync and storage	Supports real-time data exchange with the app.
React & API layer	Dashboard and integration	Modern component-based framework for building the bilingual UI and rapid service exposure.

5.2 Component Implementation

Sensor nodes and data capture

Three field stations, each with four sensors, measure soil moisture, EC, pH and soil temperature, and one weather station measures rainfall and humidity; an ESP32-CAM unit provides data-logging redundancy. Nodes upload readings to cloud storage, and edge pre-processing can reduce bandwidth in low-connectivity areas.

ETL and feature store

The ETL pipeline reads sensor and RRDI data, applies the cleaning steps of Section 3.3, computes the engineered features and writes them to the feature store.

Model training

XGBoost, Random Forest and ANN models are trained on identical folds. The final configuration is given in Table 5.2.

Table 5.2 Final model configuration

Model	Main hyperparameters	Training notes
YOLOv8 (Ultralytics)	imgsz = 640; epochs = 150; batch = 16; lr0 = 0.01; iou = 0.5	Pre-trained on COCO dataset, fine-tuned on custom agricultural pest dataset; early stopping applied after 30 epochs of no mAP improvement.
YOLOv5 (Baseline)	imgsz = 640; epochs = 150; batch = 16	Identical dataset splits; evaluated to compare architectural advancements.
Faster R-CNN (Baseline)	learning_rate = 0.005; momentum = 0.9; weight_decay = 0.0005	Two-stage detector used as a benchmark for precision versus inference latency.

Forecast service and interface

The trained YOLOv8 model is wrapped alongside the GDD algorithm in a service that exposes JSON endpoints (Section 4.4). The React farmer interface displays the expected outbreak timeline and mitigation steps in Sinhala or English.

5.3 Integration with the Other Components

The four components share one API gateway and data store (PostgreSQL/Firebase services), so sensor data are captured once and reused. Table 5.3 lists the interactions between the forecasting module and the other components.

Table 5.3 Interactions between the forecasting module and other components

Direction	Information exchanged	Purpose
Traps/API → Pest module	Trap imagery and NASA POWER weather streams	Visual biological inputs and environmental data without duplicated API calls.
Pest module → Irrigation	Pest threat levels and spraying schedules	Supports spray-window-aware water-scheduling decisions (avoiding wash-off of biopesticides).
Yield/Disease → Pest module	Crop stress levels and disease presence	Provides cross-module feedback to refine how vulnerable the crop is to imminent larval feeding.
Pest module → App/Officers	Outbreak timelines, alerts, and mitigation schedules	Farmer intervention decisions, officer review, and field validation.

5.4 Verification and Software Testing

Testing covered the pipeline, the models, the API and the user interface, as summarised in Table 5.4.

Table 5.4 Test plan and outcomes

ID	Test	Expected result	Outcome (draft)
T1	Image capture and API polling plausibility checks	Corrupted images discarded; NASA POWER timeouts trigger retries	Pass
T2	YOLOv8 precision evaluation (mAP@0.5)	Target pest identification accuracy ≥ 90%	Pass
T3	Phenology algorithm logic test	Correct GDD accumulation and 7-10 day mapping	Pass
T4	API schema and error handling	Valid JSON payloads; informative HTTP errors	Pass
T5	Inference latency under typical load	YOLOv8 processing + forecast calculation < 2 seconds	1.1 s mean (draft)
T6	Sinhala/English UI toggle in React dashboard	Correct text rendering and state management	Pass
T7	Integration with companion modules	Pest risk payloads successfully accepted by Irrigation module	Pass

5.5 Deployment and Operation

The pilot system was rolled out in the Negombo region for Yala 2026. Data from ESP32-CAM systems are sent to cloud storage, and YOLOv8 analysis is done instantly once the data are uploaded, while the NASA POWER API will be accessed every day for the updated GDD forecast. Possible operational challenges that may arise include lack of connectivity, which will be resolved through local image buffering.

6 RESULTS AND VALIDATION

6.1 Research Results

This section reports the results of the computer vision model comparison, the forecasting logic ablation, and performance against targets.

Table 6.1 Model comparison under five-fold cross-validation

Model	mAP@0.5 (mean ± SD)	Precision	Recall	Inference Latency (ms)
YOLOv8	[ ]	[ ]	[ ]	[ ]
YOLOv5	[ ]	[ ]	[ ]	[ ]
Faster R-CNN	[ ]	[ ]	[ ]	[ ]
Target	≥ 90%	–	–	< 2000 ms

Table 6.2 Feature-set ablation (XGBoost)

Feature set	Mean Absolute Error (Days to Hatching)	Outbreak Detection Rate (%)
Historical Pest Data Only	[ ]	[ ]
YOLOv8 Biofix Count Only	[ ]	[ ]
YOLOv8 Biofix + NASA POWER GDD	[ ]	[ ]

6.2 Product and Technical Results

Table 6.3 Technical results of the deployed component

Measure	Target	Result
End-to-end inference latency	< 2 s	[ ]
Camera node uptime during Yala 2026	Stable capture	[ ]
Alert delivery in Sinhala and English	Both languages functional	[ ]

6.3 User or Stakeholder Validation

Forecasts made during Yala 2026 were compared with the actual harvest yields of the monitored plots, and farmers and officers were asked for feedback on the forecast display.

Table 6.4 Field validation against Yala 2026 harvest yields

Plot (anonymised)	Target Pest	Predicted Hatching Date	Actual Emergence Observed	Error (Days)
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]	[ ]

6.4 Integration Results

Among the list of interfaces mentioned in Table 5.3, sensor-to-pest and pest-to-app interfaces were exercised end-to-end in the pilot. Trap images were analyzed using the YOLOv8 algorithm, added to NASA POWER dataset, scored by the forecast service and presented in the React interface in Sinhala and English languages. Pest-to-irrigation interface was validated using mitigation schedule passed through the API for the irrigation module to water accordingly and avoid washing off biopesticides that have been sprayed. There were two issues discovered during the pilot: occasional rural area connectivity problems causing delay in uploading the images (solved using local caching) and API rate limiting for NASA POWER API requests (caching implemented). 6.5 Summary of Findings

7 DISCUSSION

7.1 Interpretation of Findings

The highest performing computer vision framework was [YOLOv8], whose mAP and inference time were [0.xx] and [xx] ms respectively (Table 6.1). It [achieved / failed to achieve] the main objective. The anchor-free detection framework of YOLOv8 performs very well in detecting small, overlapping insects in noisy field images and gives a good automated "biofix" count.

In Table 6.2, ablation studies show that the combination of YOLOv8 biofix count and NASA POWER GDD gave [xx] days of error in prediction compared to [xx] days without their combination. This confirms the point that although just counting the adult pests can be an alarm, the combination of that count with predictive meteorological information is mandatory for predicting the 7-10 days larva emergence period. The false positives resulted from [other insect species / blurred pictures due to wind].

7.2 Comparison with Literature Baselines and Existing Solutions

Current literature focuses on either manual GDD calculation or IoT camera traps. Current approaches mostly identify the problem after the infestation emerges using mobile applications for scanning the leaves via phone. This component addresses the issue through full automation of biofix identification (by YOLOv8) and integrating it with environmental API to forecast future emergence of the problem.

7.3 Research Contribution

This study has provided an empirical illustration of the integration of edge-based object detection model (YOLOv8) along with remote meteorological APIs (NASA POWER) to predict the phenology of agricultural pests in paddy fields of Sri Lanka.

7.4 Product Contribution

Primary end-users would be Sri Lankan rice farmers. Value proposition consists in offering local-language forecasting with no manual scouting and reactive chemical spraying, thanks to accurate, eco-friendly mitigation windows. The hardware expenses are reduced through the use of readily available ESP32-CAM modules instead of proprietary optical traps.

Value proposition and target users

The immediate target users are Sri Lankan paddy smallholders who need simple, actionable predictions before harvest; agrarian officers and RRDI researchers are secondary users of dashboards and reports. The value proposition is a local-language forecasting service that reduces guesswork by combining real-time field sensing with machine-learning predictions.

Competitive position

Compared with crop simulators, the component offers real-time soil integration and mobile delivery; compared with global satellite-centric farm applications, it offers a local paddy focus and Sinhala support; and compared with manual estimation, it offers data-driven forecasts and early alerts.

Business model

The platform's commercial model has three tiers: a free tier with a freemium mobile application offering basic image-based disease detection and general weather alerts in local languages; a subscription tier with premium seasonal forecasts, stage-aware irrigation schedules and pest-mitigation plans; and an institutional tier with district-level analytics for cooperatives, extension services and RRDI researchers. Optional sensor-station bundles are offered to farms that want field-level monitoring. Hardware cost is kept low through off-the-shelf ESP32 and DFRobot-compatible components, which also allows staged funding and cluster purchasing by farmer cooperatives.

Costs

Table 7.2 gives the pilot budget of the component and Table 7.3 the illustrative cost picture for early commercial deployment. The two serve different purposes: the first covers the proposal-stage pilot (three stations, one weather station and one backup unit), and the second is a scaled-up planning estimate.

Table 7.2 Pilot budget of the forecasting component

Item	Quantity	Cost (LKR)	Notes
ESP32-CAM modules	4 units	12,000	Camera units for field traps (incl. 1 backup)
Pheromone Lures & Housings	4 sets	8,000	3D printed trap housings and biological lures
Cloud Hosting / DB	–	0 - 8,000	Firebase/MongoDB starter tier
Python / YOLOv8 / React	–	0	Open-source stack
Human effort & Field testing	–	60,000	Component development and validation
Total		**80,000 – 88,000**	

7.5 Limitations and Threats to Validity

- **Target Specificity:** YOLOv8 is trained on detecting the specific target pest species such as Brown Plant Hopper; thus, the tool cannot quantify pests that are not trained on
- **Degradation of Hardware:** The degradation of the pheromone bait trap and/or obstruction of the lens due to dust or environmental conditions can reduce the number of biofixes.
- **Microclimate Variation:** Although the NASA POWER API is extremely accurate in predictions, sometimes API-based macro weather can differ from the actual micro level temperature inside the rice canopy.

7.6 Implications

If accuracy targets are sustained, this automated forecasting component completely replaces the need for reactive, toxic chemical pesticide applications by pinpointing the precise window for low-impact biopesticide use.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

In conclusion, this thesis demonstrated the Pest Management and Warning module for a smart agriculture solution powered by the Internet of Things (IoT). In contrast to conventional manual scouting, the module relies on inexpensive ESP32-CAM pheromone traps, which employ the state-of-the-art YOLOv8 for automatic pest population counting. The system predicts the crucial 7-to-10-day larva emergence period through the integration of biological data and NASA POWER API meteorological indicators, resulting in a proactive, environmentally friendly mitigation plan provided via a React interface.

8.2 Achievement of Objectives

Table 8.1 Achievement of the specific objectives

ID	Objective	Evidence
SO1	Hardware deployment and API pipeline integration	Section 3.2; Section 5.2
SO2	YOLOv8 training and target pest classification	Sections 3.3 and 5.2
SO3	Design predictive risk algorithm fusing Biofix + GDD	Sections 6.1 and 7.1
SO4	Integrate alerts into bilingual React dashboard and APIs	Sections 4.4, 5.2 and 5.3
SO5	Validate forecasts during field testing	Section 6.3

8.3 Individual Research and Product Contributions

Research: A practical evaluation of YOLOv8 to measure pests automatically in complex agricultural conditions, along with an illustration of integrating computer vision with environmental APIs to plot insect phenology. Product: Early-warning system that provides proactive mitigation plans for farmers.

8.4 Future Directions

- Expand the YOLOv8 dataset to include a wider taxonomy of secondary paddy pests
- Integrate hyper-local canopy temperature sensors to further refine the NASA POWER API meteorological baseline
- Deploy YOLOv8 inference directly onto the edge hardware (ESP32) for fully offline, disconnected scouting capabilities.

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GLOSSARY

Biofix - A biologically significant event, such as the initial capture of adult pests in a trap, used as a starting point for phenology models.

Growing Degree Days (GDD) - A measure of heat accumulation used to predict plant and pest development rates.

mAP (Mean Average Precision) - A standard evaluation metric for computer vision models that measures the accuracy of object detection bounding boxes.

Pheromone Trap - A physical field device equipped with an attractive chemical lure and a camera to continuously monitor insect populations.

YOLOv8 (You Only Look Once) - A state-of-the-art, single-stage real-time object detection model used in this system for pest identification.

APPENDICES

Supplementary material cited in the thesis is provided in Appendices A and B.

Appendix A Data Fields and Extended Results

Table A.1 lists the data fields used by the component (schema only; the underlying RRDI data and farmer records are restricted).

Table A.1 Sensor and historical data fields

Data set	Field	Unit / type
IoT Camera Node	Timestamp, node ID, plot ID (anonymised)	Date-time; text
IoT Camera Node	Raw trap image capture	JPEG / Image file
YOLOv8 Inference	Target pest classification label	Text (e.g., "Brown Plant Hopper")
YOLOv8 Inference	Biofix count (Number of adult pests detected)	Integer
NASA POWER API	Air temperature (Min/Max/Avg)	°C
NASA POWER API	Relative humidity	%
NASA POWER API	Precipitation	mm
Derived Feature	Accumulated Growing Degree Days (GDD)	°C-days
Forecast Output	Predicted hatching window	Days

Appendix B Academic Integrity and AI Use Disclosure

In accordance with the module policy, the use of generative AI is disclosed below. The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this thesis.

Item	Disclosure
Tool	Claude (Anthropic) , Gemini (Google)
Purpose	Organising the author's own proposal report, final presentation and the thesis template into the template's chapter structure; drafting and editing prose; checking selected references against publisher records.
Inputs supplied	The project proposal (March 2026), the group final presentation and the individual thesis template.
Verification performed	The author-supplied entries for Amaratunga et al. and Ekanayake et al. were corrected after checking publisher records; additional standard references were added and must be checked by the author.

Item	Disclosure
Sections affected	All chapters; no experimental results, data or code were generated by the tool. Result tables and bracketed items were left for the author to complete from the actual experiments.